SOCT-SAM: Stochastic-Order Constrained Transport for Referring Remote Sensing Image Segmentation

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Abstract

Referring remote sensing image segmentation requires distinguishing a specific target from visually similar instances using appearance and spatial language. Semantic correspondence alone can identify the correct category while selecting an instance that violates the described relation. We propose ROTSAM, a framework that incorporates directional relations into dense correspondence inference through Stochastic-Order Constrained Transport (SOCT). Built on a frozen SAM3 backbone with expression-conditioned low-rank adaptation, ROTSAM constructs a target-selective transport plan whose demand is calibrated to the predicted target area, while a null assignment absorbs irrelevant visual mass. Contextual anchors are localized independently to avoid imposing arbitrary area budgets on reference objects. SOCT compares target and anchor location distributions along expression-defined spatial directions and penalizes violations of their stochastic ordering. An adaptive spatial filtration uses a bounded image–language-conditioned bandwidth to compare these distributions at multiple spatial positions. With the conditional parameters fixed, the resulting inference problem is convex and is solved through dual updates interleaved with entropic transport. The inferred correspondence is shared across feature scales and reused for semantic fusion and occupancy-based geometric prompting. This formulation connects directional language, spatial correspondence, and segmentation prompting within a single inference framework, providing an explicit mechanism for discouraging relation-inconsistent grounding.

1. Introduction

The increasing availability of high-resolution satellite and aerial imagery has transformed Earth observation into a data-intensive visual understanding problem. Modern remote sensing systems support applications ranging from environmental monitoring and disaster assessment to urban analysis, yet extracting a specific object of interest from complex imagery still often requires substantial human intervention [1]. Meanwhile, advances in vision–language learning have demonstrated that natural language provides a flexible interface for specifying visual concepts and spatial relationships [7, 19]. Bridging these two directions motivates *referring remote sensing image segmentation* (RRSIS), whose goal is to identify and segment the image region described by a free-form linguistic expression [28]. Instead of segmenting every object of a predefined category, RRSIS must distinguish the particular instance intended by an expression such as "the small vehicle to the left of the bridge", making pixel-level prediction explicitly dependent on both visual appearance and linguistic context.

RRSIS is substantially more challenging than referring segmentation in natural images. Remote sensing imagery exhibits extreme scale variation, dense distributions of visually similar objects, arbitrary orientations, weak foreground–background contrast, and targets that may occupy only a few pixels [16, 28]. At the same time, referring expressions frequently contain attributes, spatial relations, and contextual anchors whose meanings cannot be resolved independently of the surrounding scene. Consequently, successful RRSIS requires more than generic vision–language fusion: the model must preserve small-object evidence, identify which linguistic cues are visually relevant, discriminate among same-class instances, and retain language guidance until the final pixel-level prediction.

Early RRSIS approaches adapt architectures developed for natural-image referring segmentation, including cross-modal attention [27], language-aware visual transformers [25], and CLIP-based segmentation [24]. Dedicated remote sensing methods have sub-

sequently introduced rotation-aware interaction [16], fine-grained cross-modal decoding [15], bidirectional alignment [13], long-term semantic guidance [18], and structurally aware reasoning [8]. These advances significantly improve vision–language interaction, but an important limitation remains: *visual adaptation, cross-modal grounding, and final mask prompting are typically treated as separate stages.* In particular, the visual representation is generally adapted independently of the current referring expression, while cross-modal interaction is subsequently used to localize the target. This separation is problematic in aerial scenes, where the same image may contain many visually similar instances and the relevant visual features depend directly on the expression being queried.

A second limitation lies in cross-modal correspondence. Conventional attention-based fusion computes local token–region affinities but does not explicitly constrain how linguistic evidence should be distributed over the spatial feature map. This distinction becomes important in RRSIS because the matching problem is highly selective: only a subset of words is visually groundable, while the referred target often occupies a very small fraction of the image. Moreover, correspondence established at a single resolution can overlook small targets or suppress their features during repeated downsampling. Although recent methods improve multi-scale interaction [15, 16], establishing globally consistent token–region correspondence across scales remains insufficiently explored.

The emergence of promptable segmentation foundation models offers a new opportunity for RRSIS. SAM [11] and SAM 2 [20] provide strong generic segmentation priors, while SAM 3 further extends promptable segmentation toward concept-level visual understanding [2]. Recent remote sensing methods therefore increasingly exploit foundation models through learned prompting and task-specific adaptation [5, 6, 21]. However, foundation-model transfer introduces a new coupling problem. The model must not only ground a detailed referring expression in a domain-shifted aerial image, but also transform that grounding into reliable prompts for the mask decoder. When localization and prompt generation are learned independently, an erroneous or diffuse grounding map can directly produce misleading geometric prompts. More fundamentally, the linguistic evidence used to identify the target is not necessarily the same evidence used to guide the final segmentation.

This work is motivated by a simple observation: *the referring expression should determine both how the image is represented and where the segmentation model is prompted.* We therefore formulate RRSIS as a language-conditioned transport problem in which linguistic tokens and visual regions establish an explicit correspondence that can be reused throughout the segmentation pipeline. Based on this principle, we propose **ROTSAM**, a framework for adapting SAM 3 to referring remote sensing image segmentation. ROTSAM first conditions visual adaptation on the input expression through a Dynamic Low-Rank Vision Transformer (DL-ViT), allowing the frozen foundation backbone to emphasise expression-relevant visual features. It then introduces Multi-Scale Optimal Transport (MS-OT) to construct token–region transport plans over multiple visual resolutions, providing explicit and globally normalised cross-modal correspondence. Crucially, rather than discarding this correspondence after feature fusion, ROTSAM reuses the deepest transport plan to generate geometric point prompts through a Transport-Based Prompt Generation (TPG) module. In this way, feature adaptation, semantic grounding, and geometric prompting are linked by the same referring expression and cross-modal correspondence.

This design addresses three complementary sources of error. First, expression-conditioned low-rank adaptation reduces the domain and instance-selection gap without fine-tuning the complete foundation model. Second, multi-scale transport explicitly allocates linguistic evidence to spatial regions and preserves fine-scale correspondence for small targets. Third, transport-derived prompting makes grounding and mask prediction consistent by generating segmentation prompts directly from the learned cross-modal correspondence. Together, these components transform SAM 3 from a generic concept segmentation model into a referring segmentation framework tailored to the geometric and semantic characteristics of remote sensing imagery.

Our main contributions are summarised as follows:

- We propose **ROTSAM**, a unified foundation-model adaptation framework for RRSIS that connects expression-conditioned visual representation, cross-modal grounding, and geometric prompting rather than treating them as independent stages.
- We introduce a **Dynamic Low-Rank Vision Transformer (DL-ViT)** that conditions low-rank visual adaptation on the current referring expression. A lightweight hypernetwork predicts expression-dependent adapter modulation, enabling target-specific visual feature extraction while keeping the SAM 3 backbone frozen.
- We develop **Multi-Scale Optimal Transport (MS-OT)** for explicit token–region alignment

across the visual feature hierarchy. The resulting transport plans provide globally constrained cross-modal correspondence while preserving fine-scale evidence required for small and spatially ambiguous targets.

- We introduce **Transport-Based Prompt Generation (TPG)**, which converts the learned transport correspondence into scale-aware geometric prompts for the SAM 3 decoder. Reusing the same correspondence for both grounding and prompting reduces the inconsistency between target localisation and final mask prediction.
- Extensive experiments on four challenging RRSIS benchmarks, including RRSIS-D, RRSIS-HR, RefSegRS, and RIS-LAD, demonstrate consistent improvements over existing referring segmentation and foundation-model-based approaches. Comprehensive ablations further validate the individual contributions and complementary behaviour of expression-conditioned adaptation, multi-scale transport, and transport-guided prompting.

2. Related Work

2.1. Referring Image Segmentation and RRSIS

Referring image segmentation (RIS) aims to generate a pixel-level mask for the object specified by a natural-language expression. Early methods mainly relied on convolutional visual encoders and recurrent language representations, followed by multimodal fusion or iterative refinement. Later approaches introduced stronger cross-modal interaction mechanisms, including self-attention, bidirectional reasoning, Transformer-based fusion, and language-aware visual encoding [10, 25, 27]. CLIP-based methods such as CRIS further improved semantic correspondence by exploiting large-scale vision–language pretraining [24]. Although these methods perform well on natural-image benchmarks, they are less effective in remote sensing scenes, where small targets, large scale variation, arbitrary orientation, dense object distributions, and weak foreground–background contrast make instance-level grounding considerably more difficult.

RRSIS extends RIS to aerial and satellite imagery and requires models to resolve both linguistic ambiguity and remote-sensing-specific visual complexity. Yuan *et al.* introduced the task and established an early benchmark for language-guided remote sensing segmentation [28]. RMSIN subsequently addressed large scale variation and object rotation through intra-scale interaction, cross-scale interaction, and adaptive rotated convolution [16]. More recent methods have focused increasingly on fine-grained cross-modal reasoning. FIANet explicitly explores fine-grained image–text alignment [12], CADFormer strengthens cross-modal alignment and decoding [15], MAFN improves multimodal feature fusion [22], SBANet introduces scale-wise bidirectional alignment [13], and LSCF maintains linguistic guidance over deeper visual stages to alleviate semantic forgetting [18]. Structurally aware and reasoning-based approaches further enhance semantic consistency [8]. Despite this progress, most existing methods still rely on visual representations that are largely fixed with respect to the current expression and predominantly use attention-based fusion for language–vision interaction. This leaves room for expression-conditioned visual adaptation and more explicit token–region correspondence.

2.2. Foundation Models for Remote Sensing and Referring Segmentation

The Segment Anything family has substantially changed the design of dense prediction systems by providing strong generic segmentation priors. SAM introduced promptable class-agnostic segmentation [11], while SAM 2 extended this capability to images and videos [20]. SAM 3 further advances promptable segmentation towards concept-level visual understanding [2]. These models are particularly attractive for remote sensing because they offer strong mask-generation capability without requiring task-specific architectures to be trained from scratch.

Several works have therefore explored adapting promptable foundation models to remote sensing imagery. SAMRS uses SAM to support large-scale remote sensing segmentation data construction [23], while RSPrompter learns task-specific prompts for remote sensing instance segmentation [4]. RemoteSAM further adapts promptable segmentation to Earth-observation imagery [26]. Within RRSIS, RSRefSeg, RS2-SAM2, semantic-localisation-guided SAM, and ReSaP combine language grounding with foundation-model-based segmentation and prompting [6, 14, 17, 21]. These approaches demonstrate the benefit of strong segmentation priors, but grounding and geometric prompting are generally learned as separate stages. As a result, errors in target localisation can propagate into the geometric prompts supplied to the segmentation decoder. ROTSAM instead couples these stages by deriving both cross-modal feature alignment and geometric prompts from a shared language–vision correspondence.

2.3. Language-Conditioned Adaptation and Cross-Modal Optimal Transport

Parameter-efficient adaptation has become an effective strategy for transferring large pretrained models without updating all backbone parameters. Low-Rank Adaptation (LoRA) introduces trainable low-rank updates to frozen weight matrices, reducing optimisation cost while preserving pretrained representations [9]. However, conventional LoRA learns a fixed adaptation after training and applies it identically to every input. This is restrictive for referring segmentation because the visual features that should be emphasised depend on the current linguistic expression. Text-conditioned adaptation addresses this limitation by generating or modulating adapter parameters from language features [3]. Following this direction, ROTSAM conditions low-rank visual adaptation directly on the referring expression so that the same frozen visual backbone can produce target-specific representations for different queries.

Accurate RRSIS also requires fine-grained correspondence between linguistic tokens and spatial regions. Existing methods commonly rely on cross-attention, gating, feature concatenation, or bidirectional fusion to establish this interaction [13, 15, 25, 27]. While effective, these mechanisms mainly model local affinities and do not explicitly formulate cross-modal matching as a constrained distribution-alignment problem. Optimal transport provides such a formulation by assigning probability mass between linguistic and visual elements under a global matching cost. This property is particularly suitable for remote sensing scenes, where only a small proportion of spatial locations may correspond to the referred target and only part of the expression may be visually grounded. ROTSAM therefore introduces Multi-Scale Optimal Transport to obtain explicit token–region correspondence across multiple feature resolutions. The resulting transport plan is further reused for geometric prompt generation, providing a direct connection between language grounding, feature alignment, and final mask prediction.

3. Method

3.1. Overview and retained visual adaptation

Given an image I and referring expression E, we predict the referred mask $\widehat{Y}$. We retain the frozen SAM3 encoders, expression-conditioned dynamic LoRA, scale projections, residual gates, and the existing decoder and confidence heads. The text encoder returns contextual token features T and a sentence feature z; the adapted visual encoder and neck return $\{X^{(s)}\}_{s=1}^{S}$. Throughout, f denotes the finest available feature grid, not the deepest stage.

We replace independent uniform matching with a target-selective correspondence measure. Target tokens receive a predicted area budget; irrelevant image mass goes to null. Contextual anchors are localized separately, avoiding the assumption that linguistic importance determines object area. Directional relations impose distributional ordering on the target correspondence through an adaptive spatial filtration. The fine plan is aggregated for feature fusion and reused for positive prompting.

The retained expression-conditioned visual adaptation is

$$V = XW_0^\top + \frac{\alpha}{r}\big[(XA^\top) \odot H_{\mathrm{LoRA}}(z)\big]B^\top. \qquad (1)$$

Here W_0 is frozen; A, B and the hypernetwork are trainable. Dynamic LoRA is supporting architecture, not the principal novelty.

3.2. Language roles and multi-scale costs

A fixed dependency parser extracts the target phrase $\mathcal{T}$ and supported direct target–anchor relations $\mathcal{R} = \{(\mathcal{A}_m, d_m)\}_{m=1}^{M}$. Target and anchor attributes belong to their respective phrases. Relation words influence contextual embeddings and identify d_m, but do not consume visual mass. Exclude padding, special tokens and function words from matching columns. Cache parser outputs and align words to subwords.

For groundable tokens, retain the original projected cosine cost:

$$C_{jk}^{(s)} = 1 - \frac{\langle X_j^{(s)} W_v^{(s)}, T_k W_t \rangle}{\|X_j^{(s)} W_v^{(s)}\|_2 \|T_k W_t\|_2}. \qquad (2)$$

Stabilize denominators. Projected and transported feature dimensions must match.

Let a_j be normalized valid fine-cell area, equal to $1/N_f$ on an unpadded regular grid. Define nonnegative geometric aggregation operators D_s with $\mathbf{1}^\top D_s = \mathbf{1}^\top$ and $D_f = I$. Sum child masses on nested grids; otherwise use normalized spatial overlap. Define

$$a^{(s)} = D_s a, \qquad \overline{C}_{\cdot k} = \sum_s w_s D_s^\top C_{\cdot k}^{(s)}, \qquad w_s \geq 0, \quad \sum_s w_s = 1. \qquad (3)$$

Start with equal weights. Cost lifting and mass aggregation are distinct operations. This is multi-scale cost fusion, not a new transport solver.

3.3. Selective target demand and independent anchor localization

Predict target area and normalized within-target weights:

$$\widehat{\rho} = \sigma(H_\rho([\mathrm{Pool}(X^{(f)}); z])), \qquad \alpha_k = \frac{\exp h_k}{\sum_{v\in\mathcal{T}} \exp h_v}. \tag{4}$$

For example, h_k is produced by a small shared head on $[T_k; z]$. Transport columns comprise $\mathcal{T}$ and null:

$$b_k = \widehat{\rho}\,\alpha_k \ (k \in \mathcal{T}), \qquad b_\varnothing = 1 - \widehat{\rho}. \tag{5}$$

Append a bounded null cost, e.g. $C^{(s)}_{j\varnothing} = 2\sigma(g^{(s)}_\varnothing([X^{(s)}_j; z]))$, with zero transported value. Context still enters target embeddings and visual adaptation.

For anchor m, average its lifted costs and construct a spatial prior:

$$c^A_{mj} = \frac{1}{|\mathcal{A}_m|}\sum_{k\in\mathcal{A}_m} \overline{C}_{jk}, \qquad r_{mj} = \frac{a_j \exp(-c^A_{mj}/\theta_A)}{\sum_i a_i \exp(-c^A_{mi}/\theta_A)}. \tag{6}$$

Use a fixed validation-selected temperature $\theta_A > 0$ initially. This is a normalized location belief, not an anchor area mask. No ground-truth anchor mask enters training or inference. The prior can be inaccurate or span multiple instances. It is computed once per forward pass and held fixed during transport inference, preventing the inner optimizer from moving the anchor merely to satisfy a relation. Gradients may still reach its features during training.

3.4. Adaptive spatial filtration

Let $r^{\mathrm{pos}}_j = (x_j, y_j) \in [0,1]^2$ be a fine-cell centre in the actual augmented image frame, with y increasing downward. Encode each supported relation as target preceding anchor on an oriented axis:

$$n_{\mathrm{left}} = (1,0), \quad n_{\mathrm{right}} = (-1,0), \quad n_{\mathrm{above}} = (0,1), \quad n_{\mathrm{below}} = (0,-1). \tag{7}$$

The parser selects these fixed directions; we do not learn directions that could reverse linguistic meaning. Normalize the projection:

$$h_{mj} = \frac{n_m^\top r^{\mathrm{pos}}_j - \ell_m}{v_m - \ell_m}, \quad \ell_m = \min_{r\in[0,1]^2} n_m^\top r, \quad v_m = \max_{r\in[0,1]^2} n_m^\top r. \tag{8}$$

A small head predicts a shared target–anchor bandwidth:

$$\tau_m = \tau_{\min} + (\tau_{\max} - \tau_{\min})\sigma(H_\tau([z; z_T; z_{A_m}; \mathrm{Pool}(X^{(f)})])), \tag{9}$$

$$\phi_{m\ell j} = \sigma\left(\frac{u_\ell - h_{mj}}{\tau_m}\right). \tag{10}$$

Here z_T, z_{A_m} are phrase-pooled text features and u_ℓ are fixed evaluation positions. The same membership function is used for target and anchor. Since membership increases with u_ℓ, it defines a nested soft spatial filtration, not a post-hoc heatmap filter.

For an initial reproducible configuration, set $G = \max(H_f, W_f)$, $\tau_0 = 1/G$, $\tau_{\min} = \tau_0/2$, $\tau_{\max} = 2\tau_0$, and $u_\ell = \ell/G$ for $\ell = 0, \ldots, G$. These are starting settings, not validated optima. Do not initially learn evaluation positions or unconstrained relation-strength gates.

3.5. Stochastic-order constrained transport

For $P \in \mathbb{R}_+^{N_f\times(|\mathcal{T}|+1)}$, let t indicate target columns and $p(P) = Pt/\widehat{\rho}$. Feasibility imposes $P\mathbf{1} = a$ and $P^\top\mathbf{1} = b$, so $\sum_j p_j(P) = 1$. Define

$$F^T_{m\ell}(P) = \sum_j p_j(P)\phi_{m\ell j}, \quad F^A_{m\ell} = \sum_j r_{mj}\phi_{m\ell j}, \tag{11}$$

$$g_{m\ell}(P) = F^A_{m\ell} - F^T_{m\ell}(P). \tag{12}$$

The target should accumulate at least as much probability as the anchor toward the smaller side of the oriented axis. Thus $g_{m\ell} \le 0$ represents weak directional ordering.

We solve

$$\min_{P,\xi\ge 0} \ \langle P, \overline{C}\rangle + \varepsilon\sum_{j,k} P_{jk}(\log P_{jk} - 1) + \frac{\lambda_{\mathrm{ord}}}{2}\sum_{m,\ell}\omega_{m\ell}\xi^2_{m\ell} \tag{13}$$

$$\text{subject to} \quad P\mathbf{1} = a, \quad P^\top\mathbf{1} = b, \quad g_{m\ell}(P) \le \xi_{m\ell}.$$

Set $\omega_{m\ell} = 1/[M(G+1)]$ for $M > 0$. Eliminating slack gives the squared-hinge penalty $\lambda_{\mathrm{ord}}\sum_{m,\ell}\omega_{m\ell}[g_{m\ell}(P)]^2_+/2$. Slack accommodates uncertain anchors, extended objects and mismatches between the surrogate and linguistic meaning. For $M = 0$, omit ordering and use ordinary selective OT.

Conditioned on the input features, demands, anchor priors and bandwidths are fixed inside the solve. Therefore g is affine in P, and Eq. (13) is convex with a unique optimal plan for $\varepsilon > 0$ and positive active marginals. Network training is not convex.

Scope. Hard membership and all support breakpoints yield exact first-order stochastic ordering of projected cell-centre distributions. Finite sigmoid checks are a surrogate, not an exact-order certificate. Equality satisfies weak order; neither hard nor soft ordering guarantees instance identity. Above/below refer to the image plane, not depth or geographic directions. The initial module does not model inside, near, between, depth relations or ordinal selection.

3.6. Dual-conditioned Sinkhorn

Flatten (m, ℓ) to q, write $g_q(P) = d_q + \langle B_q, P\rangle$, and set

$$d_q = F^A_{m\ell}, \qquad (B_q)_{jk} = -\frac{\phi_{m\ell j} t_k}{\widehat{\rho}}, \qquad c_q = \lambda_{\mathrm{ord}} \omega_{m\ell}. \tag{14}$$

Using $c_q[g_q]_+^2/2 = \max_{\zeta_q \geq 0}\{\zeta_q g_q - \zeta_q^2/(2c_q)\}$, the primal solve for fixed ζ is standard OT with adjusted costs:

$$C^\zeta = \overline{C} + \sum_q \zeta_q B_q, \tag{15}$$

$$P(\zeta) = \mathrm{Sinkhorn}(C^\zeta, a, b), \tag{16}$$

$$\zeta_q^{(v+1)} = \left[\zeta_q^{(v)} + \eta_v \left(g_q(P(\zeta^{(v)})) - \frac{\zeta_q^{(v)}}{c_q}\right)\right]_+ . \tag{17}$$

Initialize $\zeta = 0$. Use FP32 log-domain Sinkhorn and warm-start its scaling variables. Choose step sizes by dual backtracking or a justified bound. This is projected ascent on a concave dual with accurate inner solves; approximate solves require controlled errors. Recompute P after the final dual update. An unchanged single Sinkhorn call does not solve Eq. (13).

Start profiling with 10 outer updates and up to 50 inner iterations per update; these are provisional budgets. Validate marginal residuals and the dual fixed-point residual $\|\zeta - [\zeta + \eta(g - \zeta/c)]_+\|_\infty/\eta$. Nonzero g can be optimal because slack is permitted. A useful dual value is

$$\mathcal{D}(\zeta) = \min_{P \in \Pi(a,b)} \{\langle P, C^\zeta\rangle + \varepsilon \sum_{jk} P_{jk}(\log P_{jk} - 1)\} + \sum_q \zeta_q d_q - \sum_q \frac{\zeta_q^2}{2c_q}. \tag{18}$$

With an accurate feasible inner solve, compare this against the primal objective to monitor the gap. Avoid presenting values from unconverged inner solves as certified bounds.

Differentiate through the implemented fixed-budget unrolling during training, including demands, anchor priors and bandwidths. A finite unrolling is approximate differentiable inference. Inference may use a validated residual-based stopping rule. Evaluate conditioning for tiny $\widehat{\rho}$ because B_q contains $1/\widehat{\rho}$.

3.7. Feature fusion, occupancy prompts and decoding

Aggregate the fine plan:

$$P^{(s)} = D_s P^*, \qquad q^{(s)} = \mathrm{diag}(a^{(s)})^{-1} P^{(s)} t. \tag{19}$$

Occupancy $q_j^{(s)} \in [0, 1]$ is the fraction of cell mass assigned to the target. It differs from normalized location probability p_j, used for ordering.

Reuse the residual gate with target-only values:

$$A^{(s)} = \mathrm{diag}(a^{(s)})^{-1} P_{\mathcal{T}}^{(s)} T_{\mathcal{T}}, \quad \widetilde{X}^{(s)} = X^{(s)} + \sigma(\mathrm{MLP}([X^{(s)}; A^{(s)}])) \odot A \tag{20}$$

The null value is zero. Anchor semantics influence inference through the order constraints and contextual embeddings, without being assigned a competing area budget.

Select spatially separated maxima of $q^{(f)}$ as positive points, with the original cap $K = 5$. Do not use $\max_k P_{jk}$ or min–max normalization. Initially retain one point when $\widehat{\rho} < 0.01$, tuning this rule on validation only. This remaining prompt-count threshold is separate from adaptive filtration. Map cell centres through actual crop/resize transforms into prompt coordinates.

Retain the existing geometry encoder, decoder interfaces and candidate selection:

$$\{(\widehat{Y}_n, s_n)\}_{n=1}^{N_q} = \mathcal{D}_{\mathrm{SAM3}}(\{\widetilde{X}^{(s)}\}, T, \mathrm{points}(q^{(f)})), \qquad \widehat{Y} = \widehat{Y}_{\arg\max_n s_n} \tag{21}$$

Use the actual existing decoder API; the notation does not establish an undocumented interface. Hard point selection supplies no ordinary gradients through selected coordinates.

3.8. Training

Compute target area ρ^* and area-averaged fine labels y_j from the augmented mask on valid support. Retain the current segmentation and global contrastive losses. Remove the Gram-matrix structural loss from the proposed full model rather than claiming it encodes direction.

$$\mathcal{L}_{\mathrm{occ}} = \mathcal{L}_{\mathrm{balBCE}}(q^{(f)}, y) + 1 - \frac{2\sum_j a_j q_j^{(f)} y_j + \delta}{\sum_j a_j q_j^{(f)} + \sum_j a_j y_j + \delta}, \tag{22}$$

$$\mathcal{L}_\rho = \mathrm{Huber}\left(\log(\widehat{\rho} + \delta_\rho) - \log(\rho^* + \delta_\rho)\right), \tag{23}$$

$$\mathcal{L}_\tau = \frac{1}{M}\sum_m [\log(\tau_m/\tau_0)]^2, \tag{24}$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{seg}} + \lambda_{\mathrm{con}}\mathcal{L}_{\mathrm{con}} + \lambda_{\mathrm{occ}}\mathcal{L}_{\mathrm{occ}} + \lambda_\rho \mathcal{L}_\rho + \lambda_\tau \mathcal{L}_\tau. \tag{25}$$

For $M = 0$, set $\mathcal{L}_\tau = 0$. Let δ_ρ be one valid image pixel's area; use a small numerical δ. Define

$$\mathcal{L}_{\text{balBCE}} = -\frac{1}{2}\left[\frac{\sum_j a_j y_j \log(q_j+\delta)}{\sum_j a_j y_j + \delta} + \frac{\sum_j a_j (1-y_j)\log(1-q_j+\delta)}{\sum_j a_j (1-y_j)+\delta}\right]. \tag{26}$$

The order penalty is already in inference and affects training through P^*; do not count it again as an outer loss initially. Bandwidth regularization discourages weakening the constraint through excessive smoothing. Fix relation strength instead of learning an unconstrained gate. Ground-truth masks, areas and occupancy labels are training-only inputs.

4. Supplementary properties and limitations

Proposition 1 (Area conservation and induced scale consistency)**.** *For feasible P^* and positive valid cell areas,*

$$0 \le q_j^{(s)} \le 1, \quad (a^{(s)})^\top q^{(s)} = \widehat{\rho}, \quad q^{(s)} = \operatorname{diag}(a^{(s)})^{-1} D_s \operatorname{diag}(a) q^{(f)}. \tag{27}$$

Proof. Target mass is bounded by row mass. Column-sum preservation of D_s preserves total target demand. Substitute $P^* t = \operatorname{diag}(a) q^{(f)}$ for the final identity. □

These are consequences of construction, not a new OT theorem; they concern occupancy rather than decoded masks.

Proposition 2 (Convex conditional inference)**.** *For fixed positive active marginals, fixed anchor maps and bandwidths, $\varepsilon > 0$ and nonnegative order strength, relaxed inference has a unique optimal plan.*

Proof. The transport polytope is nonempty and compact. Each g_q is affine; its squared positive part is convex. Entropy is strictly convex in P, giving existence and uniqueness. □

Exact stochastic ordering implies ordered means of projected positions, but ordered means do not imply stochastic ordering. Exact ordering is preserved by common strictly increasing coordinate transformations, and horizontal reflection with left/right inversion preserves the directional construction. Neither statement establishes whole-network equivariance. Finite sigmoid checks can miss hard-CDF violations.

Failures include inaccurate or multimodal anchor priors, parser mistakes, extended/overlapping objects, weak equality solutions, tiny-area conditioning and unsupported relations. Independently localizing anchors avoids forcing them to consume an arbitrary fraction of image area, but does not solve anchor identity. Distributional ordering is a geometric surrogate, not a universal definition of spatial language. No persistent homology is computed.

5. Exact changes to the attached manuscript

Original location	Required change
Title / abstract	Use the proposed title after validation. Replace the template abstract. Insert performance only after rerunning.
Introduction, lines 093–161	Make directional constrained inference the principal contribution. Treat adapters, multi-scale fusion and prompting as supporting components. Remove four-benchmark claims unless four full evaluations exist.
Section 3.1	Replace the overview with target-selective transport, independent anchor priors and adaptive order constraints. Identify the finest grid explicitly.
Section 3.2 / Eq. (2)	Keep dynamic LoRA; reduce novelty claims about conditional adaptation.
Section 3.3 / Eq. (3)	Keep cosine costs; add role parsing, null, area demand, anchor priors and cost lifting.
Section 3.3 / Eq. (4)	Replace independent uniform OT with Eq. (13); specify dual-conditioned Sinkhorn.
Section 3.3 / Eq. (5)	Keep the gate; use target-only transported values, zero null values and valid-area normalization.
Section 3.3 / Eq. (6)	Remove Gram SCL from the full model; retain only as an ablation. It does not encode direction.
Section 3.4 / Eq. (7)	Replace deepest-grid maximum-token confidence and min–max normalization with finest-grid target occupancy and separated peaks.
Section 3.5 / Eq. (8)	Keep decoder and confidence-based candidate selection. Verify actual inputs and coordinate transforms.
Section 3.6 / Eq. (9)	Replace total training loss with Eq. (25).
Section 3.6 / Eq. (10)	Keep segmentation losses for the first matched experiment.
Section 3.7 / implementation	Add area, null and bandwidth heads. Recompute parameters/runtime. Report inner/outer solver budgets and residuals instead of asserting 10 iterations suffice. Point selection remains deterministic, but its upstream maps now include new trainable heads.
Tables 1–5	Label original values Original ROTSAM baseline. Rerun the new method and ablations. Recompute metrics from per-image IoU: Table 2's threshold precisions imply a mean-IoU lower bound around 55.72%, exceeding its stated 55.64%. Check sample sets and aggregation.
Section 4.6	Add anchor/parser failures, finite soft-order approximation, equality, numerical conditioning and unsupported relations. Do not assert these are solved.

6. Implementation sequence and experimental controls

1. Save the original checkpoint, configuration, splits and per-image outputs. No source code was attached; these are module specifications, not verified repository filenames.
2. Cache target/anchor parsing and subword alignment. Use a deterministic whole-content target fallback with no ordering for failed parses; report coverage and fallback rate.
3. Implement valid areas, D_s, shared costs, area/null heads and selective transport without ordering. Verify D_s column sums and cost/plan adjoint identity.
4. Compute independent anchor priors and inspect them on an audited single-anchor subset. Do not use anchor ground truth during inference.
5. Implement hard filtration, then bounded adaptive sigmoid bandwidth. Keep directions and evaluation positions fixed. Update left/right or above/below labels under flips. Compute coordinates after all transforms.
6. Avoid relation-changing target-aware crops: disable such crops on the directional subset in every matched control unless anchor visibility can be established and invalid constraints skipped. Target masks alone do not establish anchor visibility.
7. Implement the dual solver and log marginal/fixed-point residuals. Verify gradients to costs, area, bandwidths and anchor features. Use log-domain numerical handling without a material lower floor on target area.
8. Replace prompts with occupancy peaks and correct coordinates. Retain decoder policy and match fine-grid prompting across controls.
9. Add area/occupancy supervision and bandwidth regularization. Retain the existing optimizer schedule initially; tune weights and solver budgets on validation only.

Variant	Purpose
B0	Original ROTSAM with original SCL and prompting.
B1	Target/null shared transport, area/occupancy supervision, independent anchor priors and fine prompts; no order term.
B2	B1 plus centroid relation penalty with matched inference-refinement budget.
B3	B1 plus ordinary relative-position cost using identical anchor priors.
B4	B1 plus hard-CDF ordering inside inference.
B5	B1 plus fixed-bandwidth soft filtration inside inference.
B6	B1 plus bounded learned bandwidth and regularization: proposed full model.
C1	B1 plus identical soft-order penalty as an outer training loss only.
C2	Matched attention localization with identical heads, supervision, anchor priors and prompts.
C3	Independent scale plans with matched losses, ordering and finest-grid prompting, isolating shared-scale construction.
C4	Full proposal with original Gram SCL restored as an auxiliary loss.

Run at least three seeds for B1/B5/B6 and the strongest control. Evaluate whole benchmarks and a predefined manually audited directional subset. Report mIoU/oIoU/Pr@X, point-hit rate, wrong-instance errors, tiny-target strata, bandwidth saturation, parser coverage, marginal/dual residuals, runtime and memory. Measure hard-CDF violations independently; small soft violation is insufficient.

Use synthetic layouts with known spatial relations to verify behavior. Horizontal reflection plus left/right inversion preserves the transformed target label; changing left to right without reflecting selects another target only where a valid alternative is annotated. Do not assign original masks to arbitrary counterfactual text. Evaluation-only anchor annotations on a small subset can diagnose priors, with their role clearly disclosed.

7. Contribution and related-work boundaries

Replacement contribution paragraph. We introduce stochastic-order constrained transport for directional referring segmentation. The method compares target and anchor location distributions through an expression-oriented spatial filtration and incorporates their ordering into correspondence inference. A bounded image–language-conditioned bandwidth adapts filtration softness while retaining a convex conditional transport problem. The target correspondence is shared across feature scales, semantic fusion and geometric prompting.

Positioning. Stochastic dominance, constrained optimization, entropic OT and filtration are established concepts. The candidate contribution is their integration into directional dense grounding. Priority and performance require full-paper comparison with nearest methods. Do not claim unrestricted relational reasoning, full-network equivariance, persistent homology or exact order certification from finite sigmoid checks. The replacement also differs from the earlier target/context/null proposal by localizing anchors independently; use matched controls to isolate this change.

8. Add following to Bib

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